

*na jikar kar, na fikar kar
apne jazbaton ki, tu bas qadar kar
Jo tera hai, wo tujhe mil hi jaayega
aapne bari ka tu bas sabar kar*

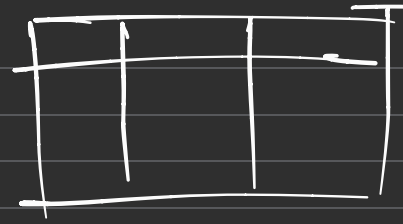
DATA BASE
SQL

SQL

- Structured Query Language
- Mainly used to interact with Database.
- Mainly Classified into three parts, i.e DDL, DML, DCL

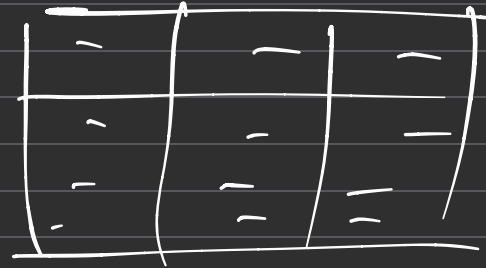
DDL (Data Definition Language)

- Mainly used to perform operation on Table.
- CREATE, TRUNACATE, DROP, ALTER



DML (Data Manupilation Language)

- Mainly used to perform operation on data of table.
- INSERT, DELETE, UPDATE, SELECT



-	-	-
-	-	-
-	-	-
-	-	-

DCL (Data Control Language)

- Used to save Permanently.
- ROLLBACK, SAVEPOINT, COMMIT, GRANT, REVOKE

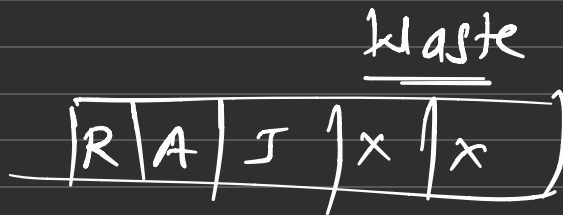
DATA TYPES IN SQL

- `number(size) / int (size)`

Mainly used to assign decimal values upto the size.

- `char(size)` $\bar{S}$

Mainly used to assign char values upto the size.



- `varchar(size)`

Mainly used to assign char values upto the size, but if string is less than the given size, it automatically get adjusted. It saves the space.



Data Definition Language

- **CREATE**

Mainly used to create the table.

* Syntax

```
CREATE TABLE <Name> (col, DT, col2 DT)
```

Eg:

```
CREATE TABLE Student (roll int, name varchar(20),  
marks int)
```

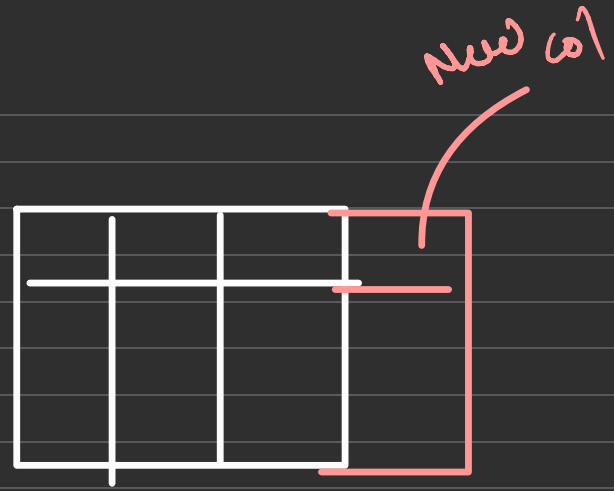
- ALTER

Mainly used when we want to add, delete any column or want to change the constraints.

Roll	Name	Marks	dept	Clg

Student

* How to add column



ALTER TABLE <Table Name> ADD <column Name> Data Type

* How to delete column

dept

please delete column

ALTER TABLE student DROP <column Name>

- DROP

Deletes the whole structure of table.

DROP TABLE <Table Name>

- TRUNCATE

Deletes all the rows. (Only delete the data)

TRUNCATE TABLE <Table Name>

Data Manipulation Language

- **INSERT**

Used to insert the data in the table.

int	varchar	int
Roll	Name	Marks

INSERT INTO <T.Name> VALUES (101, 'Yash', 70)

- UPDATE

Used to update the data or name of column in the table.

UPDATE student SET marks = 70

UPDATE student SET marks = 90
WHERE roll = 130

Roll	Name	mark
130	Sushant	90
—	—	—
150	Sushant	100

Student

update 70

- **DELETE**

Mainly delete all the content, but structure of the table remains, so in future we can still insert the data in the same table.

- **SELECT**

Mainly used to retrieve the data from the table.

Syntax:

SELECT * FROM TABLE_NAME

all columns

SELECT col1, col2 FROM Table_Name

Aggregate Functions (VVIMP)

All the aggregate functions are mainly applied on column. All the functions ignores the NULL value.

- MIN()

Returns the min value from the column.

- MAX()

Returns the max value from the column.

- SUM()

Returns the sum of all the values from the column. Must be on Integer column.

- AVG()

Returns the avg of all the values from the column. Must be on Integer column.

- COUNT()

Returns the number of rows.

	C ₁	C ₂	C ₃
	—	—	—
R ₁	—	Null	—
R ₂	—	—	—

SELECT MIN(name) FROM Student

SELECT MAX(marks) FROM Student

SELECT SUM(marks) FROM Student

SELECT AVG(marks) FROM Student

SELECT COUNT(roll) FROM Student

* MIN()

SELECT MIN(MARKS) from
student

* Sum

SELECT SUM(marks) from
student

Roll	Name	Marks
1	A	90
2	B	70
3	C	60
4	D	50

student

fun(x, arr)

LIKE

① % = any No. of characters

print all Names starting from x (x _)

SELECT Name from Student WHERE name like 'x%'

like {
x
x _
x _ _
x _ _ _
x _ _ _ _
x _ _ _ _ _

- Underscore _ = Single char

print name of student having second char
as x

[x] _ _

select name from student WHERE

name like '%x%' ✗

name like '_x%' ✓

Sub-Query

WAP to find name of employee having salary greater than 'Sneha'

→ `SELECT name from Emp WHERE sal > 40000`

`SELECT name from Emp WHERE sal >`

`(SELECT salary from Emp WHERE
name = 'Sneha')`

outer Query { Inner Query }

↓
first to be
Executed

Based on Result of
Inner Qwr. → this Runs

WAQ to find name of Emp having highest sal?

X Select name from Emp WHERE sal > 100000

select name from Emp WHERE sal = (
SELECT MAX(salary) from
Emp)

WAQ to find name of Emp having sal equal to karan

select name from Emp WHERE sal

= (SELECT salary from Emp
WHERE name = 'karan')

JOINS

Used to perform the cartesian product.

- ✓ CROSS JOINS
- ✓ NATURAL JOINS
- FULL OUTER JOINS
- ✓ THETA JOINS
- LEFT OUTER JOINS
- INNER JOINS
- RIGHT OUTER JOINS

Cross join
(No condition)

Roll	Name
1	A
2	B
3	C

T₁

X

dept	Clg
CS	x
Civil	y

T₂

Select * FROM <Table1>, <Table2>

Roll	Name	dept	Clg
1	A	CS	x
1	A	Civil	y
2	B	CS	x
2	B	Civil	y
3	C	CS	x
3	C	Civil	y

means
Cartesian
product

Theta join : It is a cross join with condition

$(T_1 \times T_2)$

condition = $T_1.\text{dept} = T_2.\text{dept}$

Roll	Name	dept
1	A	CS
2	B	Civil
3	C	CS

T_1

dept	Clg
CS	x
Civil	y

T_2

select $T_1.\text{roll}$, $T_1.\text{name}$

FROM $\langle T_1 \rangle$, $\langle T_2 \rangle$

WHERE $T_1.\text{dept} = T_2.\text{dept}$

<
>
!
T1

Roll	Name	dept	dept	Clg
1	A	CS	CS	x
2	B	Civil	Civil	y
3	C	CS	CS	x

INNER join (Equi join)

NATURAL JOIN

Perform equal operations on a common Attribute
between 2 or more Table

Roll	Name
1	A
2	B
3	C

dept	Clg
CS	X
Civil	Y



Roll	Name	dept	Clg
1	A	CS	X
1	A	Civil	Y
2	B	CS	X
2	B	Civil	Y
3	C	CS	X
3	C	Civil	Y

When no common attribute, Cartesian product

Roll	Name	dept
1	A	CS
2	B	Civil
3	C	CS

T_1

dept	Clg
CS	x
Civil	y

T_2

SELECT * from

T_1 NATURAL JOIN T_2

Roll	Name	dept	Clg
1	A	CS	x
2	B	Civil	y
3	C	CS	x

Left Outer join

(
natural join
*)

Roll	Name	dept
1	X	CS
2	Y	Civil
3	Z	CS
4	A	Mech

T₁

dept	Clg
CS	M
Civil	N
Chemical	O

T₂

Roll	Name	dept	Clg
1	X	CS	M
2	Y	Civil	N
3	Z	CS	M
4	A	Mech	Null

Select * from

T₁ LEFT JOIN T₂

Right Outer join

natural join

Roll	Name	dept
1	X	CS
2	Y	Civil
3	Z	CS
4	A	Mech

T₁

dept	Clg
CS	M
Civil	N
Chemical	O

T₂

Roll	Name	dept	Clg
1	X	CS	M
3	Z	CS	M
2	Y	Civil	N
Null	Null	Chemical	O

Select * from

T₁ RIGHT JOIN T₂

Full Outer join

(
natural join
X

Roll	Name	dept	Clg
1	X	CS	M
2	Y	Civil	N
3	Z	CS	M
4	A	Mech	Null
Null	Null	Chemical	O

Roll	Name	dept
1	X	CS
2	Y	Civil
3	Z	CS
4	A	Mech

T₁

dept	Clg
CS	M
Civil	N
Chemical	O

T₂

Select * from

T₁ FULL JOIN T₂

int 1, 2
float 3

int 1

Row →
Tuple
Record

col / Attribute / Property

Roll	name	

(name Table)

CREATE TABLE Student (roll int, name varchar(10))